

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 27

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. As we discussed in class, Amdahl's Law formalizes the theoretical speedup of a program when only part of it can be parallelized. Give a formal proof. Hint: Let T_s be execution time on 1 process, and T_p be execution time on p processes. Assume a fraction f of the program is serial and the rest is perfectly parallel. Express S_p in terms of f and p , then compute $S_p = T_s/T_p$.
2. Gustafson's Law models scaled speedup under the assumption that problem size grows with the number of processors. Give a formal proof. Hint: Assume $T_p = 1$ and split it into serial f and parallel $1-f$. Convert the parallel portion to sequential work (the total amount of parallel work) $p(1-f)$, form $T_s = f + p(1-f)$, then compute $S_p = \text{compute } S_p = T_s/T_p$.
3. Both Amdahl's Law and Gustafson-Barsis's Law are derived from the same general speedup formula. However, when increasing the number of processors p , the maximum speedup predicted by Amdahl's Law converges on $1/f$, while the speedup predicted by Gustafson-Barsis's Law increases without bound. Explain why this is so.
4. The Karp-Flatt metric is a way to estimate how much of your program is effectively serial based on observed speedup—not just theoretical assumptions. It's especially useful when real systems behave worse than ideal due to overhead. Drive an expression for the experimentally determined serial fraction. Hint: Invert speedup, then isolate the serial term.
5. The execution times of six parallel programs, labeled I-VI, have been benchmarked on 1, 2, ..., 8 processors. The following table presents the speedups achieved by these programs.

Proc	Speedup					
	I	II	III	IV	V	VI
1	1.00	1.00	1.00	1.00	1.00	1.00
2	1.67	1.89	1.89	1.96	1.74	1.94
3	2.14	2.63	2.68	2.88	2.30	2.82
4	2.50	3.23	3.39	3.67	2.74	3.65

Proc	Speedup					
	I	II	III	IV	V	VI
5	2.78	3.68	4.03	4.46	3.09	4.42
6	3.00	4.00	4.62	5.22	3.38	5.15
7	3.18	4.22	5.15	5.93	3.62	5.84
8	3.33	4.35	5.63	6.25	3.81	6.50

For each of these programs, choose the statement that best describes its likely performance on 16 processors:

- A. The speedup achieved on 16 processors will probably be at least 40 percent higher than the speedup achieved on eight processors.
 - B. The speedup achieved on 16 processors will probably be less than 40 percent higher than the speedup achieved on eight processors, due to the large serial component of the computation.
 - C. The speedup achieved on 16 processors will probably be less than 40 percent higher than the speedup achieved on eight processors, due to the increase in overhead as processors are added.
- Justify your answer.

- **Welcome: Sean**

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